

WASP-39b

R-1526

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-39_180520 · created 2026-08-01T17:39:31+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2458258.7287 +/- 0.0039 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1847 +/- 0.0092
Transit depth (Rp/Rs)^2	3.41 +/- 0.34 [%]
Orbital Inclination (inc)	85.79 +/- 0.48
Airmass coefficient 1 (a1)	1.133 +/- 0.01
Airmass coefficient 2 (a2)	-0.101 +/- 0.024
Scatter in the residuals of the lightcurve fit is	0.92 %
Best Comparison Star	#1 - [609, 385]
Optimal Aperture	3.56
Optimal Annulus	8.2
Transit Duration (day)	0.094 +/- 0.012

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1847 ± 0.0092 vs 0.1457 (deviation 2.83σ)
Duration fit vs published	0.094 d vs 0.12 d (deviation 1.45σ)
Mid-time O–C	18.2 min
Depth SNR	13.4
Residual scatter	0.92 %

Light curve

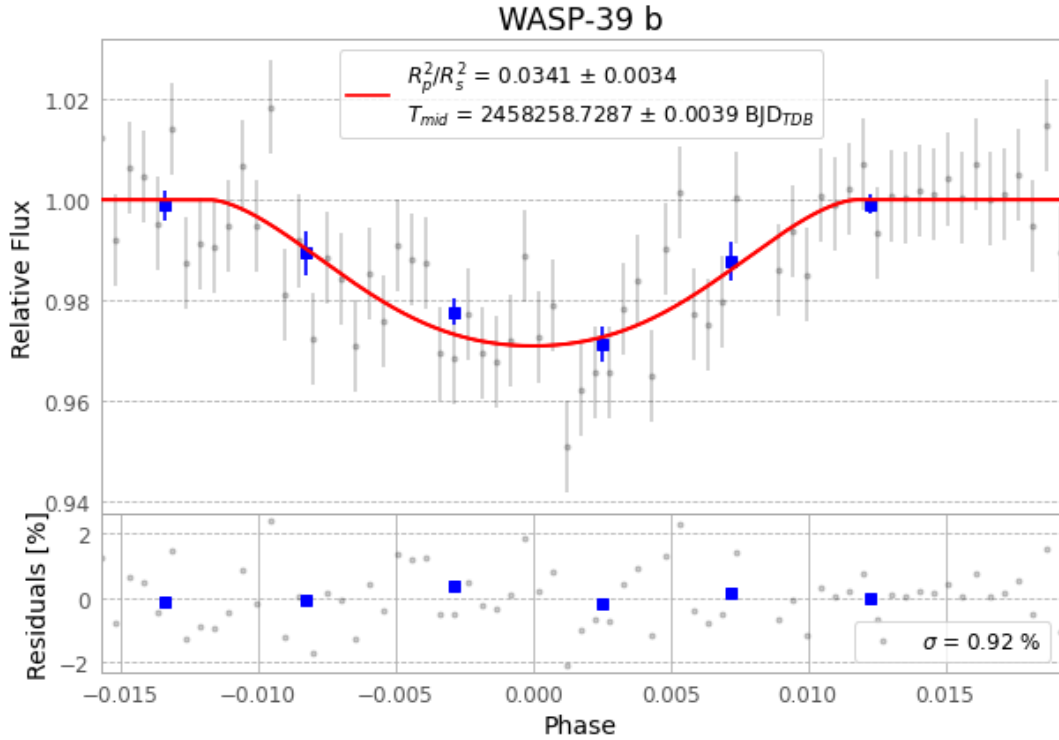


Figure 1 — FinalLightCurve_WASP-39 b_2018-05-19.png (SHA-256 17a696c4ff0618fe...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [357, 205], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 5d7cc7c6369c6f4d...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

69 FITS frames (45 MB), WASP-39180520034812.FITS → WASP-39180520071212.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-39-180520.log (93812e95e2e6...), FinalLightCurve_WASP-39 b_2018-05-19.png (17a696c4ff06...), FinalLightCurve_WASP-39 b_2018-05-19.csv (ee94aea1777c...)

Compute resources

Wall time 135.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 17:39Z.